### Step - 1 : Procedure / Implementation

```
# Install required libraries
!pip install transformers
!pip install datasets
!pip install torch
!pip install scikit-learn
```

```
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from datasets) (3.24.3)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.0.2)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
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Requirement already satisfied: tqdm>=4.66.3 in /usr/local/lib/python3.12/dist-packages (from datasets) (4.67.3)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets) (3.6.0)
Requirement already satisfied: multiprocess<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=
Requirement already satisfied: huggingface-hub>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (1.5.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from datasets) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from datasets) (6.0.3)
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2
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Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.3)
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Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0
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Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0-
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->d
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>=0.24.
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Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>=0.24.
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hu
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer-
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->type
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12
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Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
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Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
```

```
import torch
from datasets import load_dataset
from transformers import AutoTokenizer
from transformers import AutoModelForSequenceClassification
from transformers import Trainer, TrainingArguments
from sklearn.metrics import accuracy_score, precision_recall_fscore_support
```

### Step - 2 : Text Preprocessing

```
from datasets import load_dataset

# Load dataset
dataset = load_dataset("imdb")

# Create train and test variables
```

```
train_data = dataset["train"]
test_data = dataset["test"]
```

```
# Load pretrained tokenizer (DistilBERT)
tokenizer = AutoTokenizer.from_pretrained("distilbert-base-uncased")
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
```

```
config.json: 100%  483/483 [00:00<00:00, 9.86kB/s]
tokenizer_config.json: 100%  48.0/48.0 [00:00<00:00, 677B/s]
vocab.txt: 232k/? [00:00<00:00, 2.73MB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
tokenizer.json: 466k/? [00:00<00:00, 4.35MB/s]
```

### Step - 3 : Vocabulary Creation & Encoding

```
def tokenize_function(example):

    # Tokenize text
    return tokenizer(
        example["text"],
        padding="max_length",
        truncation=True,
        max_length=256
    )

# Apply tokenization to dataset
tokenized_train = train_data.map(tokenize_function, batched=True)
tokenized_test = test_data.map(tokenize_function, batched=True)
```

```
Map: 100%  20000/20000 [00:31<00:00, 803.51 examples/s]
Map: 100%  5000/5000 [00:05<00:00, 938.18 examples/s]
```

### Step - 4 : Train-Test Split

```
dataset = load_dataset("imdb")
dataset = dataset["train"].train_test_split(test_size=0.2)
train_data = dataset["train"]
test_data = dataset["test"]
```

```
README.md: 7.81k/? [00:00<00:00, 163kB/s]
plain_text/train-00000-of-00001.parquet: 100%  21.0M/21.0M [00:01<00:00, 129MB/s]
plain_text/test-00000-of-00001.parquet: 100%  20.5M/20.5M [00:01<00:00, 62.5MB/s]
plain_text/unsupervised-00000-of-00001.p(…): 100%  42.0M/42.0M [00:01<00:00, 43.0MB/s]
Generating train split: 100%  25000/25000 [00:01<00:00, 24668.05 examples/s]
Generating test split: 100%  25000/25000 [00:01<00:00, 22557.84 examples/s]
Generating unsupervised split: 100%  50000/50000 [00:01<00:00, 45658.79 examples/s]
```

### Step - 5 : Create PyTorch Dataset Format

```
tokenized_train.set_format(
    type="torch",
    columns=["input_ids", "attention_mask", "label"]
)

tokenized_test.set_format(
    type="torch",
    columns=["input_ids", "attention_mask", "label"]
)
```

### Step - 6 : Load Transformer Model

```
model = AutoModelForSequenceClassification.from_pretrained(
    "distilbert-base-uncased",
    num_labels=2   # positive / negative
)
```

```
model.safetensors:  96%                                                    256M/268M  [00:06<00:00,  68.3MB/s]

Loading weights: 100%                          100/100  [00:00<00:00,  321.67it/s,  Materializing  param=distilbert.transformer.layer.5.sa_layer_norm.weight]

DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                      | Status     |
-------------------------+------------+-
vocab_transform.bias     | UNEXPECTED |
vocab_transform.weight   | UNEXPECTED |
vocab_layer_norm.bias    | UNEXPECTED |
vocab_layer_norm.weight  | UNEXPECTED |
vocab_projector.bias     | UNEXPECTED |
classifier.weight        | MISSING    |
pre_classifier.weight    | MISSING    |
pre_classifier.bias      | MISSING    |
classifier.bias          | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING   :those params were newly initialized because missing from the checkpoint. Consider training on your downstream t
```

### Step - 7 : Define Evaluation Metrics

```
def compute_metrics(pred):

    labels = pred.label_ids
    preds = pred.predictions.argmax(-1)

    precision, recall, f1, _ = precision_recall_fscore_support(
        labels,
        preds,
        average='binary'
    )

    acc = accuracy_score(labels, preds)

    return {
        "accuracy": acc,
        "f1": f1,
        "precision": precision,
        "recall": recall
    }
```

### step - 7 : Training Configuration

```
training_args = TrainingArguments(

    output_dir="./results",

    eval_strategy="epoch",

    learning_rate=2e-5,

    per_device_train_batch_size=8,

    per_device_eval_batch_size=8,

    num_train_epochs=2,

    weight_decay=0.01,

    logging_dir="./logs"
)
```

```
`logging_dir` is deprecated and will be removed in v5.2. Please set `TENSORBOARD_LOGGING_DIR` instead.
```

### Step - 8 : Model Training

```
trainer = Trainer(

    model=model,

    args=training_args,

    train_dataset=tokenized_train,
```

```
    eval_dataset=tokenized_test,

    compute_metrics=compute_metrics
)

# Train the model
trainer.train()
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as tru
  super().__init__(loader)
```

[ 144/5000 25:31 < 14:32:49, 0.09 it/s, Epoch 0.06/2]

| Epoch | Training Loss | Validation Loss |
|---|---|---|

### Step - 9 : Model Evaluation

```
results = trainer.evaluate()

print(results)
```